

ARALSK, Kazakhstan

WMO: 357460

Lat: 46.780N

Lon: 61.660E

Elev: 62

StdP: 100.58

Time Zone: 5.00 (E05)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-26.2	-23.8	-29.3	0.3	-25.8	-26.8	0.3	-23.5	15.2	-4.0	12.4	-9.5	4.5	0	0.620

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.0	38.5	20.9	36.8	20.3	35.1	19.6	22.9	34.9	21.8	33.4	20.7	32.2	4.1	50

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.7	13.7	28.9	17.5	12.6	28.2	16.2	11.6	27.6	68.1	34.9	63.9	33.4	60.2	32.2	28.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
12.7	10.9	9.6	DB	-28.8	41.4	3.4	1.7	-31.2	42.6	-33.2	43.6	-35.1	44.6	-37.5	45.8	
			WB	-29.0	23.7	3.2	1.9	-31.3	25.1	-33.2	26.2	-35.0	27.3	-37.4	28.7	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	9.2	-11.0	-9.6	0.6	12.0	19.5	25.3	27.2	25.8	18.4	9.3	-0.2	-7.6		
	DBStd	14.80	6.92	7.47	6.98	5.59	4.46	3.89	3.48	4.14	4.74	4.95	6.05	6.28		
	HDD10.0	2471	651	549	298	43	1	0	0	2	74	307	545			
	HDD18.3	4181	909	783	549	199	39	2	0	2	56	281	556	803		
	CDD10.0	2197	0	0	6	104	296	461	534	488	253	53	2	0		
	CDD18.3	866	0	0	0	10	76	213	276	233	58	2	0	0		
	CDH23.3	11236	0	0	2	117	899	2803	3757	2990	643	26	0	0		
	CDH26.7	5824	0	0	0	28	345	1482	2093	1623	250	3	0	0		
(14) Wind	WSAvg	4.3	4.4	4.7	5.1	5.0	4.7	4.3	4.1	4.1	3.7	3.9	3.9	4.2		
(15) Precipitation	PrecAvg	144	11	9	14	15	15	14	10	7	7	14	13	13		
	PrecMax	267	23	31	41	53	60	105	59	44	24	83	40	33		
	PrecMin	91	0	0	2	0	0	1	0	0	0	0	0	2		
	PrecStd	40	7	8	10	14	14	21	13	9	8	16	9	9		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	2.3	7.9	20.9	29.7	34.7	39.9	41.1	40.1	36.4	26.2	15.7	4.8		
		MCWB	1.3	6.3	11.6	16.2	19.4	20.4	21.7	22.4	18.6	15.0	10.3	2.7		
	2%	DB	1.1	3.5	15.8	26.7	32.5	37.5	38.6	38.0	31.9	23.1	11.8	2.8		
		MCWB	0.5	2.0	9.2	15.2	18.2	20.4	20.9	21.1	17.6	13.0	8.1	1.4		
	5%	DB	0.1	1.5	12.7	23.8	30.5	35.6	36.8	36.1	29.4	20.5	9.4	1.1		
		MCWB	-0.3	0.4	7.6	13.5	17.2	19.5	20.4	20.1	16.5	11.9	6.5	0.1		
	10%	DB	-1.1	0.2	9.7	21.1	28.4	33.7	35.0	34.2	27.0	17.8	7.3	0.1		
		MCWB	-1.7	-0.6	5.7	12.3	16.2	18.8	19.8	19.1	15.7	10.5	5.0	-0.8		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	1.6	6.1	12.3	17.3	21.1	23.4	24.6	24.3	20.6	15.9	11.5	3.3		
		MCDB	2.0	7.7	19.8	27.4	33.0	35.9	36.9	38.3	32.8	24.1	15.4	4.4		
	2%	WB	0.6	2.1	9.8	15.7	19.5	22.0	23.1	22.7	18.8	14.3	8.7	1.6		
		MCDB	1.0	3.2	14.9	25.1	30.6	34.1	34.6	35.4	29.4	20.8	11.4	2.4		
	5%	WB	-0.3	0.5	7.9	14.4	18.1	20.9	21.9	21.2	17.4	12.7	6.7	0.3		
		MCDB	0.1	1.3	12.2	22.2	28.0	32.6	32.7	33.3	27.9	18.8	9.1	1.0		
	10%	WB	-1.6	-0.5	6.2	12.9	16.9	19.9	20.9	19.9	16.1	11.2	5.0	-0.7		
		MCDB	-1.1	0.2	9.5	20.1	26.3	31.1	31.9	31.8	25.9	17.0	7.1	0.0		
(35) Mean Daily Temperature Range	MDBR		6.9	8.1	8.8	11.2	12.2	12.4	12.0	12.5	12.4	11.2	7.7	6.5		
	5% DB	MCDBR	5.4	6.9	12.6	14.3	14.4	14.5	13.8	14.5	14.9	14.8	9.9	6.3		
		MCWBR	5.1	5.5	7.4	7.0	6.6	6.1	5.7	6.3	7.1	8.3	6.6	5.4		
	5% WB	MCDBR	5.1	6.6	11.8	12.9	12.9	13.0	12.5	13.3	13.6	13.1	9.1	5.5		
		MCWBR	4.9	5.7	7.3	6.8	6.7	6.4	6.4	6.8	7.3	7.8	6.5	5.1		
(40) Clear-Sky Solar Irradiance	taub		0.325	0.348	0.421	0.454	0.455	0.440	0.425	0.411	0.372	0.369	0.353	0.331		
	taud		2.073	2.029	1.989	1.964	2.036	2.145	2.210	2.226	2.284	2.203	2.153	2.097		
	Ebn at Noon		720	788	782	798	814	829	836	832	834	763	684	659		
	Edh at Noon		96	124	151	170	164	148	137	129	111	103	89	84		
(44) All-Sky Solar Radiation	RadAvg		1.32	2.29	3.45	5.16	6.71	7.17	6.90	6.18	4.77	2.96	1.56	0.93		
	RadStd		0.27	0.54	0.34	0.39	0.40	0.43	0.29	0.27	0.17	0.21	0.15	0.15		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (2 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page